

Generation Interconnection Request System Impact Study Report

For Queue Project Q18-0944

Canal 138 kV
October 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0944. The Interconnection Customer proposes a 164.7 MW solar photovoltaic facility interconnecting at the Canal 138 kV substation in the Pennsylvania area. The request was received on February 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 316 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,580,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Cranberry 500 kV/345 kV XFMR	Lighthouse Hill / Aes Beaver Vly Partners Beaver Valley 345 KV	109.3	728	5.1
Orchard Road / Shawville Sta 500 KV	Piney Crk Project / Hilltop 500 KV	100.0	2,560	10.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Cranberry 500 kV/345 kV XFMR	Replace transformer with higher-rated unit	\$20,416,000	13% (\$2,713,000)
Orchard Rd / Shawville Station 500 KV L	Reconductor line	\$53,437,000	4% (\$1,867,000)
	Total allocated to project		\$4,580,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.